Bisimulation Makes Analogies in Goal-Conditioned Reinforcement Learning

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Introduction

Objective:

Goal-conditioned RL has the potential to train agents that can accomplish a variety of tasks when given a goal. Prior works tends to use the desired state as a goal, but this is not always optimal. In this work, we study a better representation for goals. One that captures *functional equivariance* between tasks while remaining invariant to information not changeable by the agent.

Contributions:

We propose Goal-Conditioned Bisimulation (GCB), our contributions are as follows

1. an objective that learns a functionally equivariant abstraction over goal conditioned tasks
2. a demonstration that this embedding space is capable of composing states with task abstractions to depict new goals
3. a novel value bound that can be applied to any downstream task
4. an evaluation on manipulation environments that shows improved performance on goal-conditioned tasks compared to other self-supervised representation learning methods

Background

Goal-Conditioned Reinforcement Learning:

We assume that our problem setting follows a goal-conditioned markov decision process. Importantly, the reward function is sparse, meaning an agent gets a reward if 1 when the state matches the goal and 0 otherwise.

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{G}, \mathcal{R}, \gamma)$$

Bisimulation:

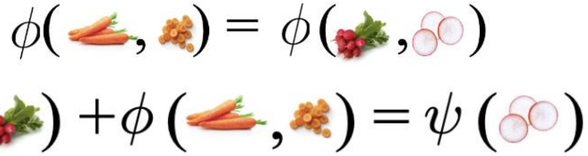
Definition 4.1 (Goal-conditioned Bisimulation Relations). Given an MDP $\mathcal{M}$, as described in section 3, we define an equivalence relation B between states. This relation is a goal-conditioned bisimulation relation if, for all state-goal pairs $(\mathbf{s}_i, \mathbf{g}_i), (\mathbf{s}_j, \mathbf{g}_j) \in \mathcal{S} \bigcup \mathcal{G}$ that are equivalent under B (denoted $\mathbf{s}_i \equiv_B \mathbf{s}_j$) the following conditions hold:

$$\begin{aligned} \mathcal{R}(\mathbf{s}_i, \mathbf{a}, \mathbf{g}_i) &= \mathcal{R}(\mathbf{s}_j, \mathbf{a}, \mathbf{g}_j), \quad \forall \mathbf{a} \in \mathcal{A}, \\ \mathcal{P}(G|\mathbf{s}_i, \mathbf{a}) &= \mathcal{P}(G|\mathbf{s}_j, \mathbf{a}) \quad \forall \mathbf{a} \in \mathcal{A}, \forall G \in \mathcal{S}_B, \end{aligned}$$

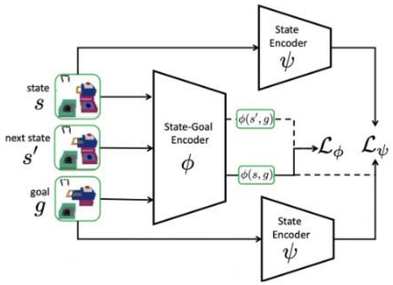
$$\begin{aligned} d_\pi(\mathbf{s}_i, \mathbf{g}_i; \mathbf{s}_j, \mathbf{g}_j) = \\ &|\mathcal{R}(\mathbf{s}_i, \pi(\mathbf{s}_i, \mathbf{g}_i), \mathbf{g}_i) - \mathcal{R}(\mathbf{s}_j, \pi(\mathbf{s}_j, \mathbf{g}_j), \mathbf{g}_j)| \\ &+ \gamma \mathcal{W}_2(\mathcal{P}(\mathbf{s}'_i|\mathbf{s}_i, \pi(\mathbf{s}_i, \mathbf{g}_i), \mathcal{P}(\mathbf{s}'_j|\mathbf{s}_j, \pi(\mathbf{s}_j, \mathbf{g}_j)) \end{aligned}$$

Single State Representation:

- Need a way to compose new goals from known task abstractions
- Single state representation should match following arithmetic



Method



Algorithm 1 GCB Training Algorithm

Given: RL algorithm $\mathcal{A}$, replay buffer $\mathcal{D} = \{(s, a, s', r, g)_i\}_{i=1}^N$, learning rates α_ψ, α_ϕ
Initialize π, ϕ, ψ
for training iteration $k = 1, 2, \ldots$ **do**
  $B_1 \leftarrow \{s_i, a_i, s'_i, r_i, g_i\}_{i=1}^B \sim \mathcal{D}\{\text{Sample Batch}\}$
  $B_2 \leftarrow \{s_j, a_j, s'_j, r_j, g_j\}_{j=1}^B = \text{permute } B_1$
  $\phi \leftarrow \phi - \alpha_\phi \nabla_\phi \mathcal{L}_\phi(B_1, B_2)$
  $\psi \leftarrow \psi - \alpha_\psi \nabla_\psi \mathcal{L}_\psi(B_1)$
  Update $\pi(\psi(s), \phi(s, g))$ with $\mathcal{A}$ on B_1
end for
for eval episode $m = 1, 2, \ldots$ **do**
  Sample analogous tasks (s, g) and (s_a, g_a)
  Rollout policy with $\pi(\psi(s), \phi(s_a, g_a))$
  Compute success of final achieved state against g
end for

$$\mathcal{L}_\phi = \Big(||\phi(s_i, g_i) - \phi(s_j, g_j)||_1 - ||r_i - r_j||_2 - \gamma ||\bar{\phi}(s'_i, g_i) - \bar{\phi}(s'_j, g_j)||_2 \Big)^2,$$

$$\mathcal{L}_\psi = \Big((\bar{\phi}(s_i, g_i) - \bar{\phi}(g_i, g_i)) - (\psi(g_i) - \psi(s_i)) \Big)^2$$

Train:

Test:

Value Bounds and Sufficiency Results

Proposition 6.2. *For any two state goal pairs* $(\mathbf{s}_i, \mathbf{g}_i), (\mathbf{s}_j, \mathbf{g}_j) \in (\mathcal{S}, \mathcal{G})$ *and a given policy* π,

$$|V^\pi(\mathbf{s}_i, \mathbf{g}_i) - V^\pi(\mathbf{s}_j, \mathbf{g}_j)| \leq d^\pi(\mathbf{s}_i, \mathbf{g}_i; \mathbf{s}_j, \mathbf{g}_j). \quad (9)$$

In other words, if two tasks have small distance in our metric, the policy will exhibit similar performance on both tasks.

Proposition 6.5. *For any reward function* $R \in \mathcal{R}$ *as defined above and a given policy* π,

$$|V_R^\pi(\mathbf{s}_i) - V_R^\pi(\mathbf{s}_j)| \leq \sum_{k=0}^{|\mathcal{S}|} \alpha_k d^\pi(\mathbf{s}_i, \mathbf{g}_k; \mathbf{s}_j, \mathbf{g}_k), \quad (10)$$

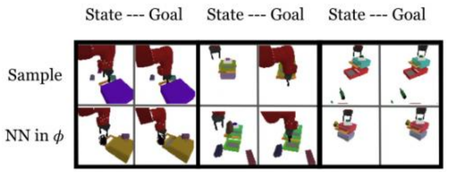
where V_R^π *denotes the value function for a given policy* π *and reward function* R.

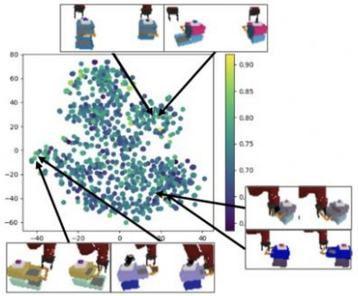
Which shows that such a metric is *universal* for any downstream task with *any* potential reward function.

Visualizations and Experimental Results

GCB Nearest Neighbors:

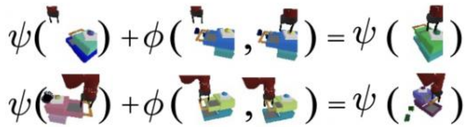
- Nearest Neighbors of Task Space
- TSNE of Task Space





Analogy Arithmetic Visualized:

- Nearest Neighbors of Analogy Arithmetic
- State + Analogous Task = Inferred Goal



Manipulation Results:

- Run Offline RL (IQL) + Representation Learning Methods
- Analogy Experiment train with exact goal, but use analogous task as goal at test time
- Also run standard GCRL where exact goal is always provided

Figure 4. `Button and Drawer` env with video distractors.

Method	`Drawer`	`Drawer+VD`	`BD`	`BD+VD`	`Analogy`	`Analogy+VD`
GCB (ours)	**0.533 ± 0.037**	**0.448 ± 0.033**	0.440 ± 0.026	**0.322 ± 0.021**	**0.403 ± 0.041**	**0.303 ± 0.027**
CPV	0.400 ± 0.031	0.209 ± 0.013	0.228 ± 0.024	0.198 ± 0.013	0.176 ± 0.011	0.121 ± 0.009
GC-DBC	0.269 ± 0.025	0.261 ± 0.031	0.358 ± 0.023	0.255 ± 0.034	NA	NA
CCVAE	0.443 ± 0.029	0.231 ± 0.017	0.452 ± 0.039	0.225 ± 0.028	0.234 ± 0.012	0.095 ± 0.009
VAE	0.463 ± 0.023	0.261 ± 0.018	**0.522 ± 0.042**	0.115 ± 0.019	0.280 ± 0.024	0.112 ± 0.006
CPC	0.305 ± 0.020	0.207 ± 0.014	0.363 ± 0.042	0.154 ± 0.023	0.148 ± 0.026	0.145 ± 0.029
CURL	0.259 ± 0.021	0.283 ± 0.030	0.382 ± 0.038	0.198 ± 0.022	0.162 ± 0.021	0.123 ± 0.025
RAD	0.245 ± 0.019	0.207 ± 0.010	0.159 ± 0.036	0.108 ± 0.018	0.157 ± 0.028	0.110 ± 0.026
Pixel	0.483 ± 0.051	0.256 ± 0.027	0.117 ± 0.016	0.136 ± 0.027	0.132 ± 0.025	0.121 ± 0.033

Table 1. Final avg. success rates with stderr over 5 seeds. `BD` refers to the `Button and Drawer` env. and `VD` refers to Video Distractors.